

4.2 Forces: Density & Pressure

Question Paper

Course	CIE A Level Physics
Section	4. Forces, Density & Pressure
Topic	4.2 Forces: Density & Pressure
Difficulty	Easy

Time allowed: 40

Score: /33

Percentage: /100

Question 1a

State the word definition of density.

[1 mark]

Question 1b

When calculating density both the equation and units are required.

(i)

State the equation for density, defining all the terms used.

[1]

(ii)

State the SI units for density and a non-SI unit which is commonly used.

[2]

[3 marks]

Question 1c

For pressure caused by a solid, state

(i)

the definition of pressure

[1]

(ii)

the equation for pressure, defining the terms used.

[1]

[2 marks]

Question 1d

Fig. 1.1 shows a measuring cylinder containing a liquid with depth h and cross-sectional area A .

Using the equations you stated in (b) and (c), show that pressure at a point in the liquid is $\Delta P = \rho g \Delta h$

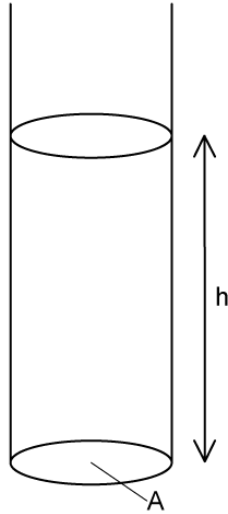


Fig. 1.1.

[2 marks]

Question 2a

The water in the measuring cylinder shown in Fig 1.1. has a depth h of 10 cm and a cross-sectional area A of 1.0 cm^2 .

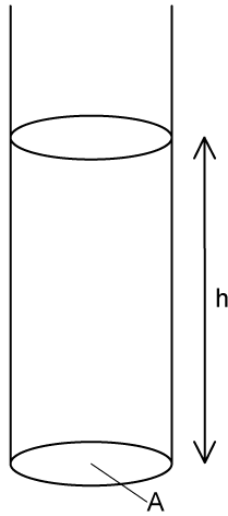


Fig 1.1.

- (i)
State the measurements of h and A in S.I. units.

[2]

- (ii)
Calculate the pressure at the base of the measuring cylinder due to the water. The density of water = 1000 kg m^{-3} .

[2]

- (iii)
Calculate the total pressure at the base of the measuring cylinder. Atmospheric pressure = $1.01 \times 10^{-5} \text{ Pa}$.

[1]

[5 marks]

Question 2b

The water pressure is measured at the base of two other measuring cylinders of different sizes.

One of the cylinders contains water with a depth five times larger than the cylinder in (a).

The other cylinder contains the same depth of water as the cylinder in (a), but the cross-sectional area of the base is nine times larger.

Without further calculation, state the effect on the water pressure at the base of the cylinder where

(i)

the water is five times deeper

[1]

(ii)

the surface area of the base is nine times larger.

[1]

[2 marks]

Question 2c

A spherical ball is placed in the measuring cylinder so that it is submerged but floats in the water, as shown in Fig. 1.2.

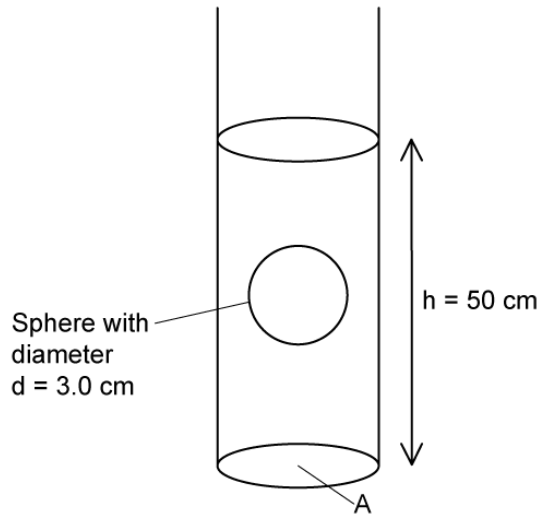


Fig. 1.2.

The ball has a diameter of 3.0 cm .

Calculate the volume of the ball in S.I. units.

[3 marks]

Question 2d

Calculate the upthrust on the submerged ball in part (c).

[3 marks]

Question 3a

Complete the definition of hydrostatic pressure.

Hydrostatic pressure is the pressure that is exerted by a _____ at _____ at a given point within the fluid, due to the force of _____.

[3 marks]

Question 3b

A cylinder of length 1.8 m is submerged vertically in a tank of seawater so that the top of the cylinder is 0.2 m below the surface of the water, as shown in Fig. 1.1.

The density of the seawater is 1020 kg m^{-3} .

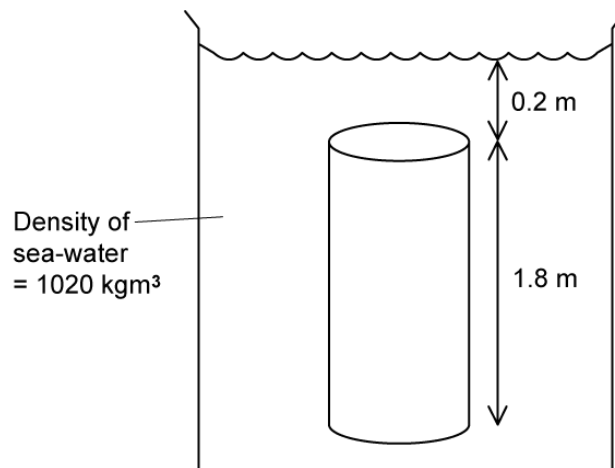


Fig. 1.1.

Calculate the difference in hydrostatic pressure between the top and bottom surfaces of the cylinder.

[2 marks]

Question 3c

The cylinder has a diameter of 75 cm.

Calculate the upthrust on the cylinder.

[4 marks]

Question 3d

The seawater in Fig. 1.1. is replaced with pure water which has a lower density than seawater.

Without further calculation

(i)

State and explain the effect this would have on the upthrust on the cylinder

[2]

(ii)

Hence, state whether the cylinder would sink, float or stay at the same position in the tank.

[1]

[3 marks]



Model Answers: Easy

1a

(a) State the word definition of density:

- The mass per unit volume; [1 mark]

[Total: 1 mark]

1b

(b)

(i) The equation for density is:

- $\rho = \frac{m}{V}$ where ρ = density, m = mass and V = volume; [1 mark]

(ii) State the SI units and a common non-SI unit for density:

- SI units are kg m^{-3} ; [1 mark]
- Commonly used non-SI units are g cm^{-3} ; [1 mark]

[Total: 3 marks]

1c

(c)

(i) Pressure is defined as:

- Force per unit area; [1 mark]

(ii) The equation for pressure is:

- $P = \frac{F}{A}$ where P = pressure, F = force and A = area; [1 mark]

[Total: 2 marks]

1d

(d) Show that in a liquid the pressure at the depth h is found using $\Delta P = \rho g \Delta h$:

Write out all the equations which will be needed:

- Pressure: $P = \frac{F}{A}$ (equation 1)
- At the depth h , the force is the **weight** of liquid at h : $W = mg$ (equation 2)
- Density: $\rho = \frac{m}{V}$ (equation 3)
- Volume of a cylinder, $V = A h$ (equation 4)

Substitute *equation 3* into *equation 1* to get a new term for the weight:

- $W = \rho V g$ [1 mark]

Replace the term V with the expression in *equation 4*:

- $W = \rho A h g$ (equation 5)

Substitute equation 5 into equation 1 and cancel to complete the derivation:

- $P = \frac{F}{A} = \frac{\rho \cancel{A} h g}{\cancel{A}}$ [1 mark]
- $\Delta P = \rho g \Delta h$

[Total: 2 marks]

When doing a derivation it's helpful to write down any relevant equations before you start.

Also keep the equation you are deriving in front of you, so that you can see when you are getting close.

Finally, practice!

2a

(a)

(i) State both measurements in S.I. units:

- $h = 10 \text{ cm} = 0.1 \text{ m}$ [1 mark]

Calculate cm^3 to m^3 in steps:

- $1 \text{ cm} = 0.01 \text{ m}$
- $1 \text{ cm}^2 = 0.01^2 \text{ m}^2$
- $A = 1.0 \times 10^{-4} \text{ m}^2$ [1 mark]

(ii) Hence calculate the pressure at the base of the measuring cylinder due to the water:

- Hydrostatic pressure: $\Delta P = \rho g \Delta h$
- $P = 1000 \times 9.81 \times 0.1$ [1 mark]
- $P = 981 = 980 \text{ Pa}$ [1 mark]

(iii) Calculate the total pressure at the base of the measuring cylinder:

- Total pressure = pressure caused by the water + atmospheric pressure
- $P = 980 + (1.01 \times 10^5) = 1.02 \times 10^5 \text{ Pa}$ [1 mark]

[Total: 5 marks]

2b

(b)

(i) If the water became five times deeper:

- Pressure and depth are directly proportional, so...
- (The pressure would be) five times larger; [1 mark]

(ii) If the surface area became nine times larger:

- The depth is the same, so...
- (The pressure would be) the same; [1 mark]

[Total: 2 marks]

The instruction 'without further calculation' is a clue that something straightforward will get you the answer. In this case, pressure is directly proportional to depth, but since area isn't in the equation for pressure in a liquid, it has no effect and can be ignored.

2c

(c) Calculate the volume of the ball in S.I. units:

Convert the diameter to S.I. units:

- diameter, $d = 3.0 \text{ cm} = 0.030 \text{ m}$
OR
- radius, $r = 0.015 \text{ m}$ [1 mark]

Replace radius, r with diameter, d in the equation for the volume of a sphere:

- Volume of a sphere: $V = \frac{4}{3} \pi r^3$
- $V = \frac{4}{3} \pi (0.015)^3$ [1 mark]
- $V = 1.41 \times 10^{-5} \text{ m}^3$ [1 mark]

[Total: 3 marks]

2d

(d) Calculate the upthrust on the submerged ball in part (c):

Recall the equation for upthrust:

- Upthrust: $F = \rho g V$ [1 mark]

- Where ρ and V are the density and volume of the displaced fluid

Substitute known quantities:

- Density of water, $\rho = 1000 \text{ kg m}^{-3}$ (from **(a)**)
- Volume, $V = 1.41 \times 10^{-5} \text{ m}^3$ (from **(c)**)

Substitute the known quantities and calculate:

- $F = 1000 \times 9.81 \times (1.41 \times 10^{-5})$ [1 mark]
- $F = 0.14 \text{ N}$ (2 s.f.) [1 mark]

[Total: 3 marks]

3a

(a) Complete the definition of hydrostatic pressure:

- Hydrostatic pressure is the pressure that is exerted by a **fluid**; [1 mark]
- at **equilibrium** at a given point within the fluid; [1 mark]
- due to the force of **gravity**; [1 mark]

[Total: 3 marks]

3b

(b) Calculate the difference in hydrostatic pressure between the top and bottom surfaces of the cylinder:

List the known quantities:

- Density of seawater, $\rho = 1020 \text{ kg m}^{-3}$
- Height of cylinder, $\Delta h = 1.8 \text{ m}$
- Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$

Use the equation for hydrostatic pressure:

- Hydrostatic pressure: $\Delta P = \rho g \Delta h$
- $\Delta P = 1020 \times 9.81 \times 1.8$ [1 mark]

- $\Delta P = 18\,000\text{ Pa}$ (2 s.f.) [1 mark]

[Total: 2 marks]

3c

(c) Calculate the upthrust on the cylinder:

List the known quantities:

- Radius of cylinder, $r = 75\text{ cm} \div 2 = 37.5\text{ cm} = 0.375\text{ m}$
- Density of seawater, $\rho = 1020\text{ kg m}^{-3}$

Calculate the volume, V of the cylinder to find the volume of displaced sea water:

- Volume of a cylinder = height $\times$ cross-sectional area (of a circle)
- Volume of a cylinder: $V = \pi r^2 h$
- $V = \pi \times (0.375)^2 \times 1.8$ [1 mark]
- $V = 0.795 = 0.80\text{ m}^3$ [1 mark]

Substitute values for ρ and V into the equation for upthrust and calculate:

- Archimedes' Principle (upthrust): $F = \rho g V$
- $F = 1020 \times 9.81 \times 0.795$ [1 mark]
- $F = 7955 = 8000\text{ N}$ (2 s.f.) [1 mark]

[Total: 4 marks]

3d

(d)

(i) If the density of the liquid decreased:

- Upthrust would be less; [1 mark]

Calculate the volume, V of the cylinder to find the volume of displaced sea water:

- Volume of a cylinder = height $\times$ cross-sectional area (of a circle)
- Volume of a cylinder: $V = \pi r^2 h$
- $V = \pi \times (0.375)^2 \times 1.8$ [1 mark]
- $V = 0.795 = 0.80 \text{ m}^3$ [1 mark]

Substitute values for ρ and V into the equation for upthrust and calculate:

- Archimedes' Principle (upthrust): $F = \rho g V$
- $F = 1020 \times 9.81 \times 0.795$ [1 mark]
- $F = 7955 = 8000 \text{ N}$ (2 s.f.) [1 mark]

[Total: 4 marks]

3d

(d)

(i) If the density of the liquid decreased:

- Upthrust would be less; [1 mark]
- (Because) upthrust is proportional to density (of the displaced fluid); [1 mark]

(ii) The cylinder would:

- Sink lower / float at a deeper depth
- OR**
- The distance between the cylinder and surface of the water would increase / be more than 0.2 m (**NOT** the cylinder would sink to the bottom); [1 mark]

[Total: 3 marks]

Be careful of guessing more than you can know from the information given.

We can be sure the cylinder would sink to a lower position, but not that it would rest on the bottom of the tank.